

9 Observed Outcomes and the Data Boundary

The required propositions derived planning-stage and observed-product outcomes. This section fixes their interpretation and timing. It adds no primitive, control, realization channel, market, or empirical identification claim.

9.1 Outcome hierarchy

The baseline preserves the strict conceptual distinction

$$\boxed{\text{upstream research} \neq \text{project advancement} \neq \text{commercialization organization} \neq \text{realized product.}} \quad (1)$$

Here x_i is project-development / advancement intensity. It advances viable projects toward commercialization-relevant route planning and excludes patent applications, basic research and upstream scientific discovery. The optimal planning-stage arrival intensity is $\Lambda_i^{\text{plan}} = a_i x_i^*$; route choice then assigns a realized project to I, E, T , or A ; retained holder–producer separation can then be observed, followed by downstream product realization.

9.2 Anticipated and realized sequences

The ex ante policy channel is

$$M \longrightarrow \text{anticipated availability and value of } E \longrightarrow \Omega_i \longrightarrow x_i^*. \quad (2)$$

The later realized sequence is

$$x_i^* \longrightarrow \Lambda_i^{\text{plan}} \longrightarrow r_i^* \longrightarrow \text{observed holder–producer separation} \longrightarrow \text{realized products.} \quad (3)$$

Observed separation is therefore a downstream route outcome, not an event that precedes or causes the ex ante advancement choice.

9.3 Novelty classes under one common control

For $g \in \{O, \text{Inc}\}$, the class-specific retained outcome retains the form established in Proposition 6:

$$Y_{ig}^{\text{ret}} = a_i x_i^* \rho_g \int s_g(q) [\mathbf{1}\{r_i^* = I\} + \mathbf{1}\{r_i^* = E\}] dF_g(q, m). \quad (4)$$

Here $s_g(q) = s(q)$, as in Proposition 6. The classifier changes the conditional distribution used in the integral; it does not create a control x_{ig} . The baseline permits either class to have a zero response while the other responds, and imposes no ranking between their reform gains. As Proposition 6 explains, a zero conditional value gain alone does not imply zero observed change when common advancement rises.

9.4 Measurement boundary

Patent applications measure a patenting margin that may reflect upstream research activity, but they are not direct observations of upstream research and are not baseline endogenous outcomes. An isolated patent decline does not by itself falsify the mechanism. Activity around an IND or another regulatory application, and early clinical milestones, may proxy project advancement only when their measurement definition and timing are validated. Product-level holder–manufacturer

separation may proxy route E only when holder identity, manufacturer identity, product, and date are aligned. Approval or launch data are realized-product outcomes, not direct observations of x_i .

Approval-side and product-side data generally do not separately identify

$$\{x_i, a_i, s(q), \tau_E, \mu_E, p_m\}. \quad (5)$$

They may discipline composite objects such as planning-stage intensity, retained-route realization, or holder–producer separation. A primitive-complete theory does not imply primitive-by-primitive empirical identification.

9.5 Scope conclusion

The formal distinction is therefore among project-development / advancement intensity, route-planning-stage arrival, commercialization organization, and observed approval, launch, or retained product. Patent generation is not added as a baseline outcome, and downstream realization probabilities remain policy invariant. No empirical data set or proxy is asserted to be available here.